

SOUND

The Nature of Sound

WHAT IS SOUND?

- Energy.
- Vibration.
- Sound is a moving/vibrating wave that transmits through solid, liquid or gas.
- Contains motion seeking equilibrium/goal.
- The search for equilibrium produces wave of energy that sets into motion air molecules at certain frequencies and loudness, enabling us to perceive sound.

HUMAN HEARING

.....

sound wave



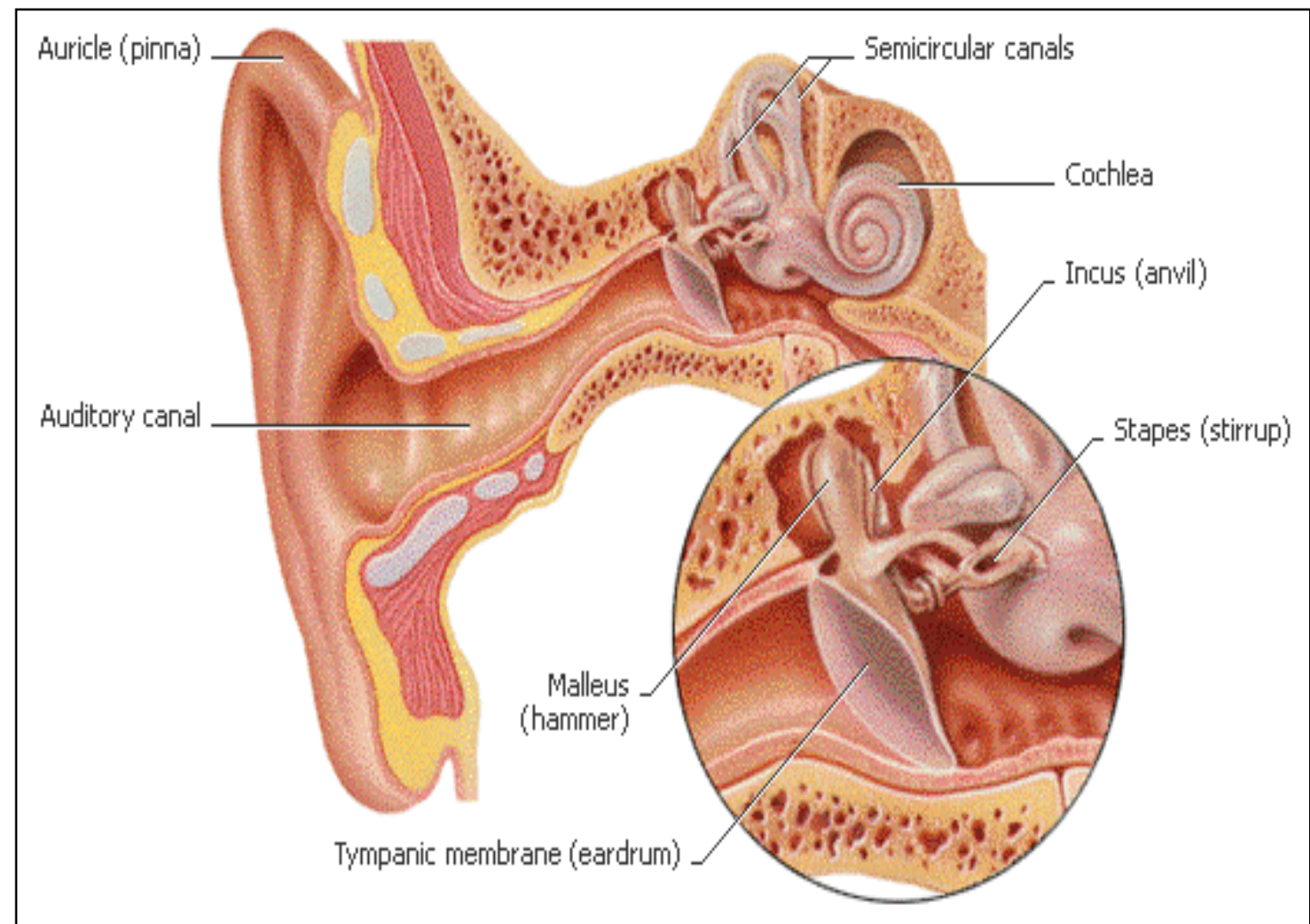
vibrates ear drum



amplified by bones



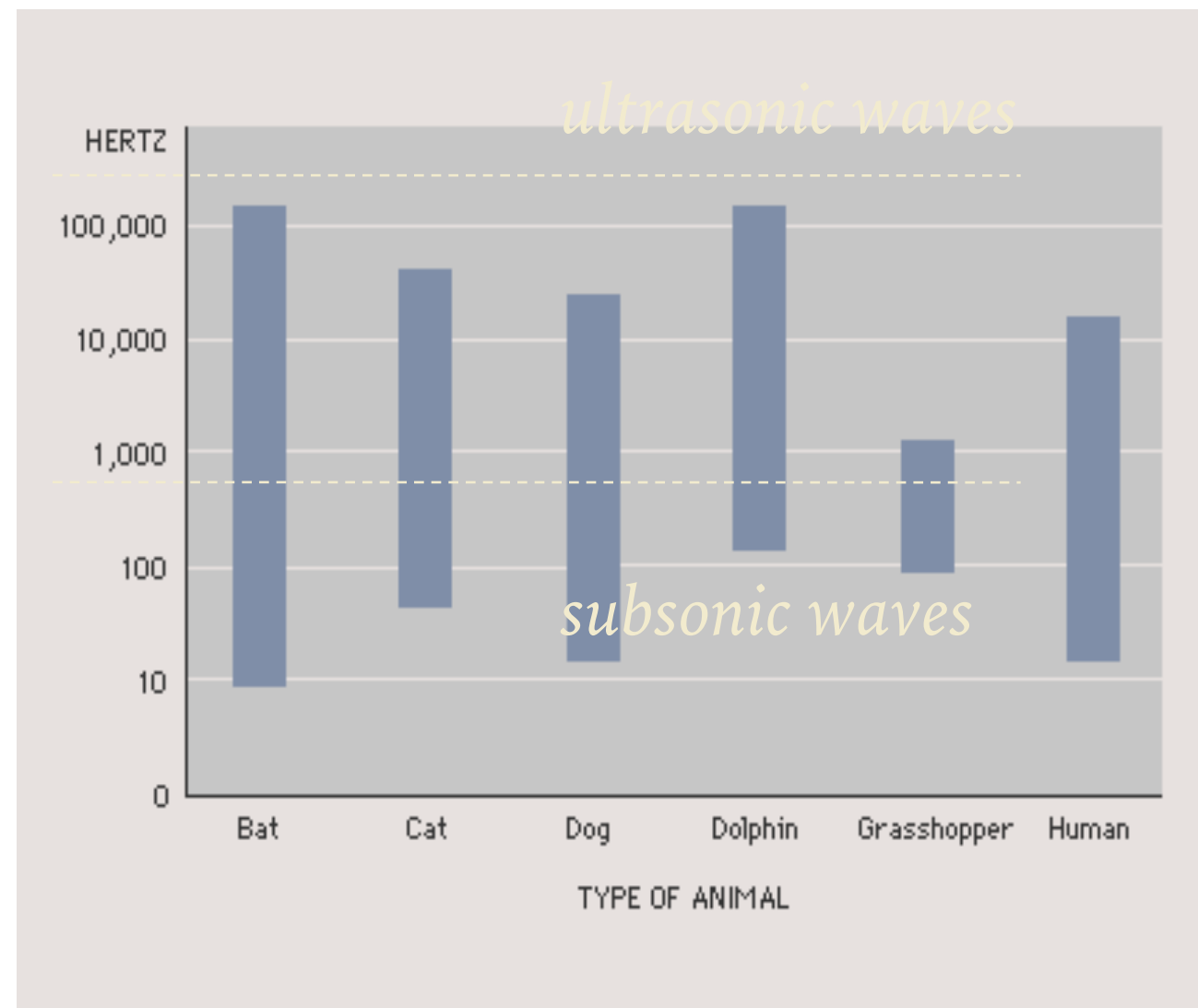
*converted to nerve impulses
in cochlea*



HUMAN HEARING

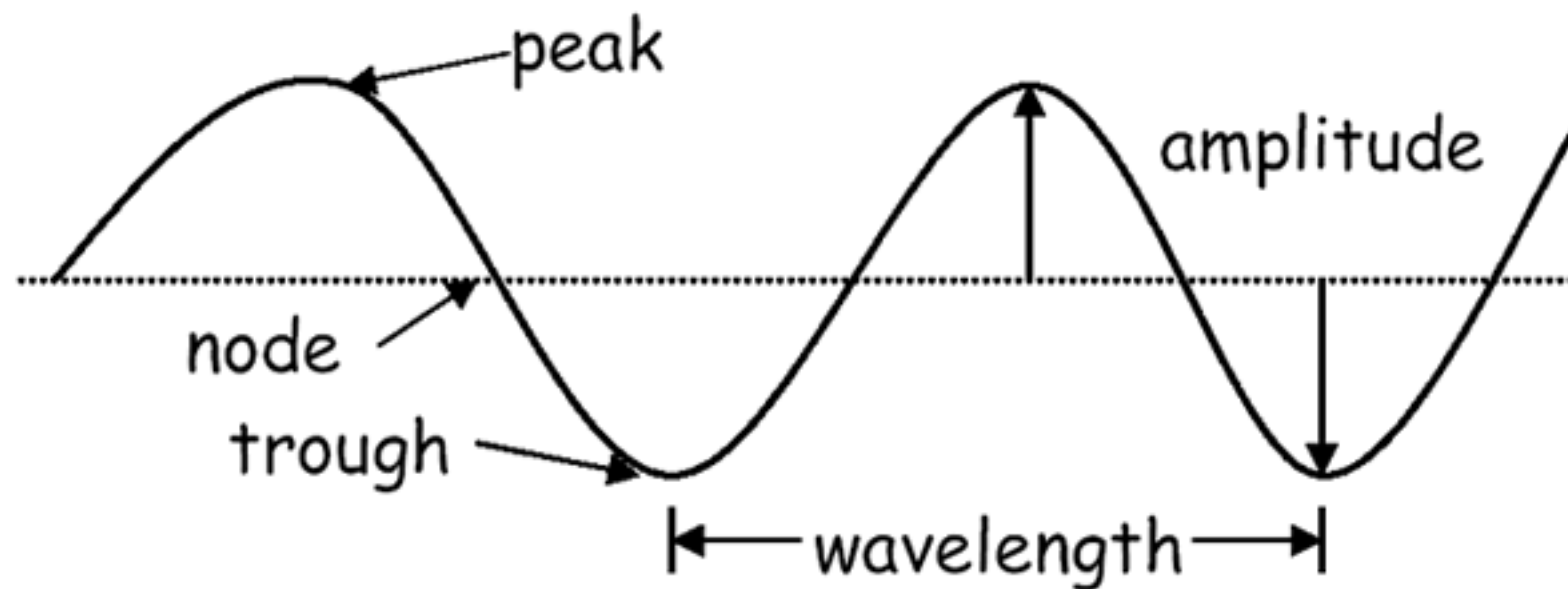
➤ Pitch

- highness or lowness of a sound
- depends on frequency of sound wave
- human range: 20 - 20,000 Hz

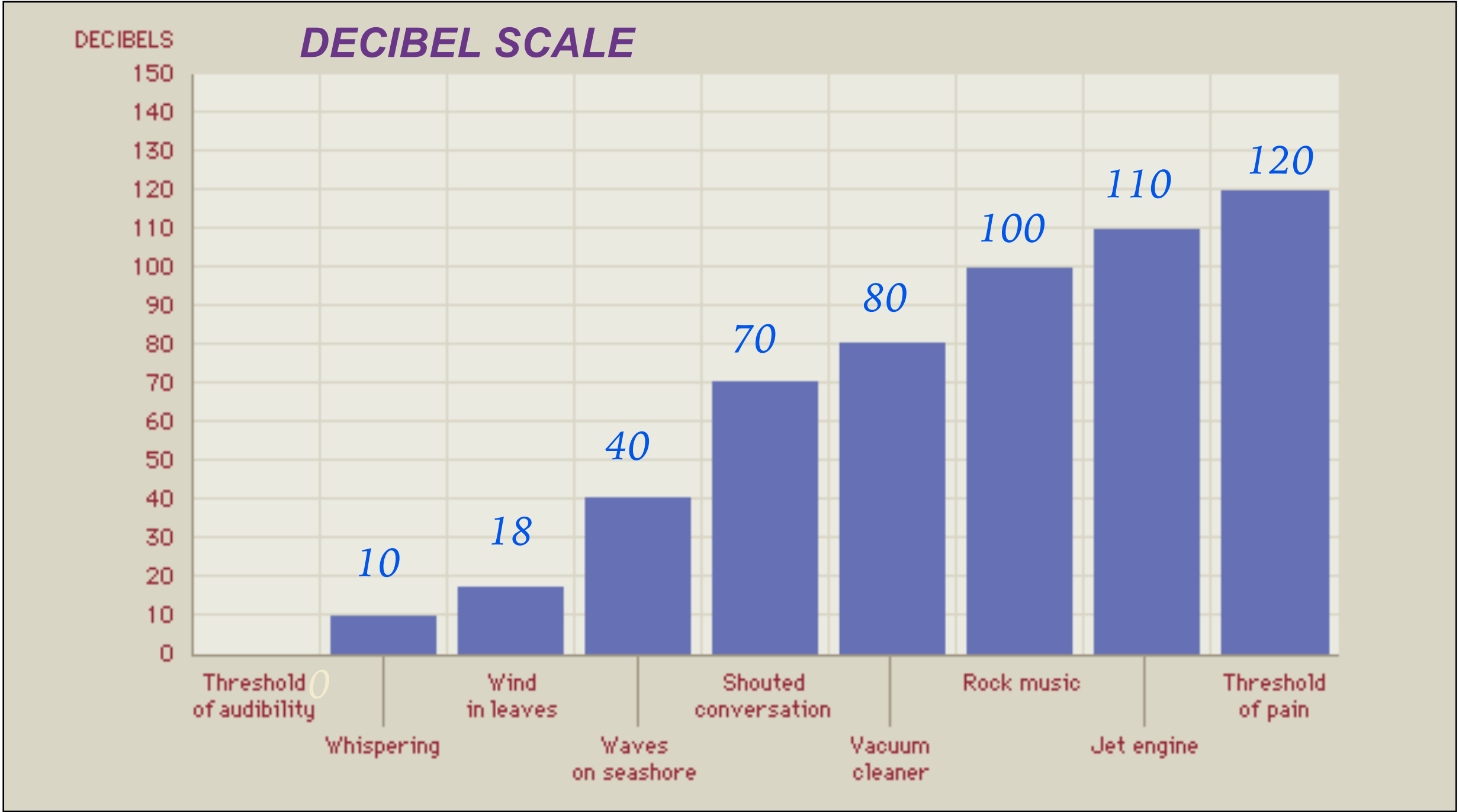


HUMAN HEARING

- Intensity
 - volume of sound
 - depends on energy (amplitude) of sound wave
 - measured in decibels (dB)



HUMAN HEARING

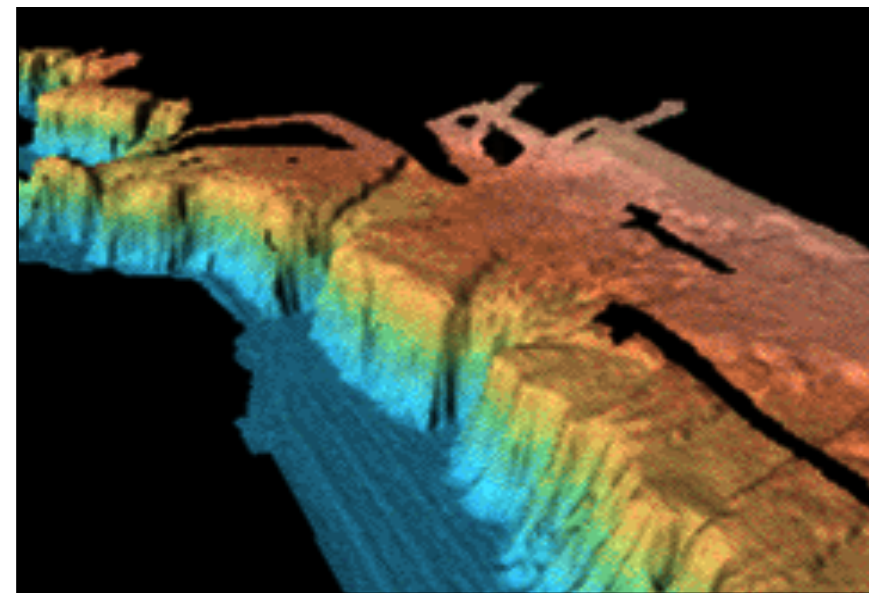


“SEEING” WITH SOUND

- Ultrasonic waves - above 20,000 Hz



Medical Imaging



SONAR

*“Sound Navigation
Ranging”*

MUSIC/SOUND VS NOISE

- Music/ Sound = specific pitches and sound quality....regular pattern
- Noise = no definite pitch...no definite pattern

40-80 HZ PINK NOISE AT -20DBFS

.....

500–2500 HZ PINK NOISE AT –20DBFS

.....

WHITE NOISE

.....

PINK NOISE

Notice the
difference?

THE PHYSICS OF SOUND

- Sound is a wave that interacts with the environment.
- **Frequency:** the number of cycles that a vibration completes in one second, measured in Hz.
- **Pitch:** the psychological perception of frequency.
- **Amplitude/Intensity:** energy/pressure of sound waves in the atmosphere, measured in dB.

THE PHYSICS OF SOUND

- **Loudness:** perceived from both amplitude and pitch.
- **Timbre:** the tone quality or “color” of a sound. Allows you to distinguish two sounds of the same pitch and loudness.
- **Shape/Envelope:** attack, initial decay, sustain, release - ADSR

AUDIO RESOLUTION

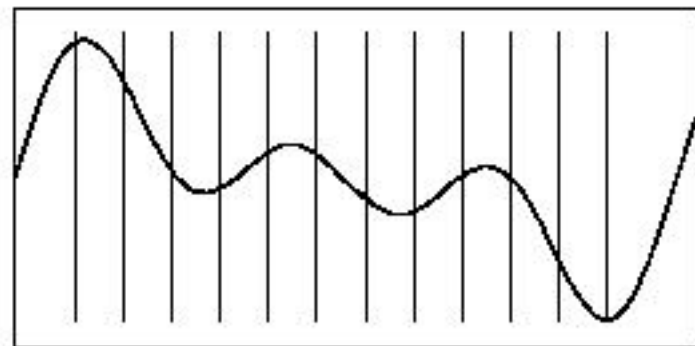
➤ NYQUIST THEOREM

The theory states that the upper threshold of a piece of digital audio will top out at half the sample rate.

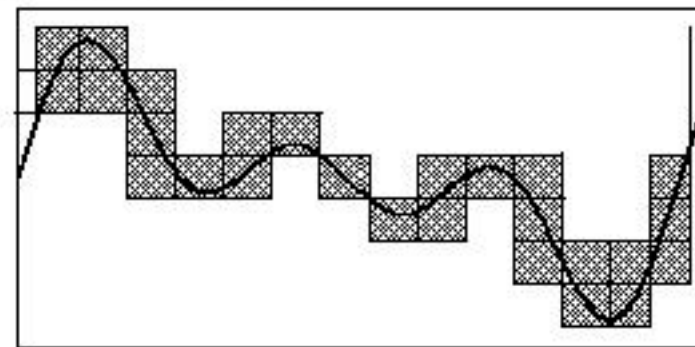
At 44.1, the highest frequencies are 22khz, 2khz higher than the typical human with excellent hearing.

SAMPLE RATE

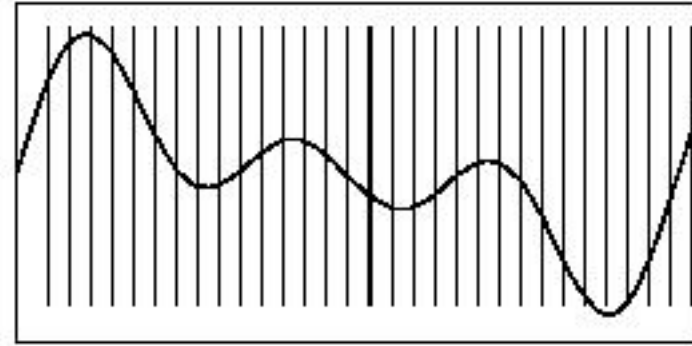
- The **SAMPLE RATE** is the number of times your audio is measured (sampled) per second.
- 96khz sample rate=96,000 slices of audio sampled each second.



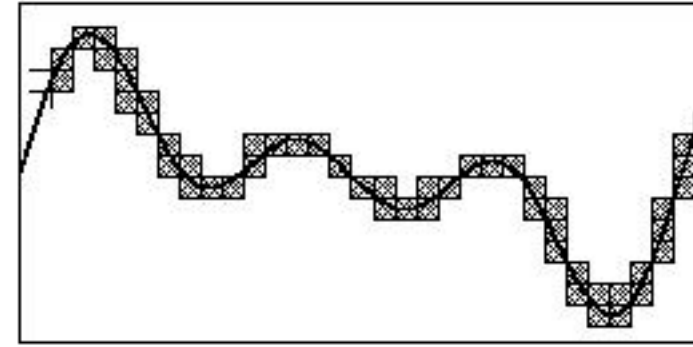
lower sample rates take fewer snapshots of the waveform



resulting in a rough recreation of the waveform.



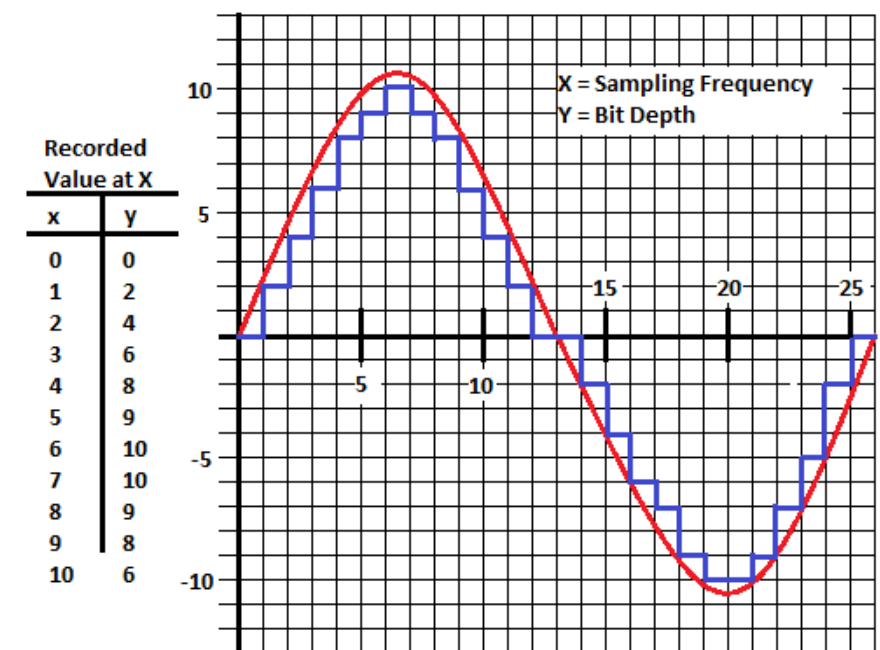
faster sample rates take more snapshots....



resulting in a smoother and more detailed recreation of the waveform.

BIT DEPTH

- Bit depth is a series of levels, a **slice of audio frozen in a single moment of time**. With 16 bit Audio, there are 65,536 possible levels.
- With every bit of greater resolution, the number of levels double. By the time we get to 24 bit, we actually have 16,777,216 levels.



RESOLUTION FACTS

- **A GREATER SAMPLE RATE ALLOWS MORE OF A SOUND TO BE SAMPLED OVER TIME.**
- **A HIGHER BIT RESOLUTION ALLOWS FOR A BETTER DYNAMIC RANGE OF THE SOUND.**
- **HIGHER BIT DEPTH AND SAMPLE RATE RESULT IN INCREASED AUDIO RESOLUTION.**

SOUND DESIGN AREAS

- Speech - dialog, voiceover, background voices
- Sound Effects (sfx) – Foley, production effects, hard effects, backgrounds
- Music – diegetic, non-diegetic, underscore, transition, psychological

SOUND FUNCTIONS

- **Contextual sound/Narrative sound** - sound interprets the visuals as it happens- (car starts, normal drive, car issues)
- **Focusing attention** (also through sound field - subject and direction)
- **Defining space** (Mood and Atmosphere)

SOUND FUNCTIONS

- **Creating environment** (Does it sound like where we are?)
- **Emphasizing/Intensifying action**
- **Setting pace as Gameplay Function** (Rhythm Games)
- **Transitions** (Game States/Mechanics)

CONTEXTUAL/NARRATIVE SOUND

Perception



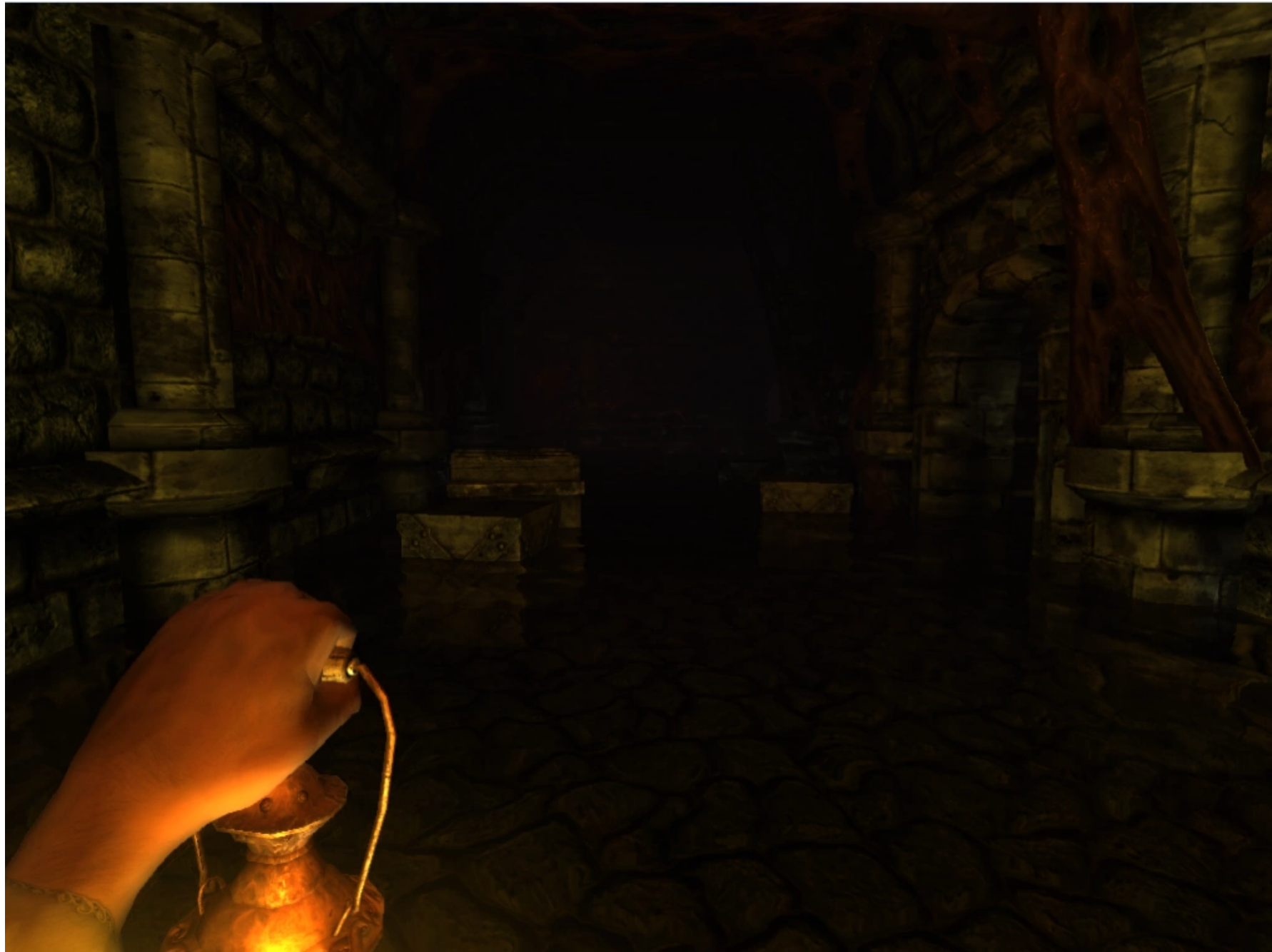
ESTABLISHING A PLACE

Bioshock



FOCUSING ATTENTION

Amnesia: The Dark Descent



CREATING ENVIRONMENT

World of Warcraft



EMPHASIZING/INTENSIFYING ACTION

DOOM



SETTING PACE as Gameplay Function

Parappa The Rapper



Gameplay and Game State TRANSITIONS

Ms. Pacman

